

Atomic scale demonstration of ferromagnetism in a single layer FeCl₂ on Au(111)

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(Dated: May 22, 2026)

FeCl₂ is a promising single-layer material with sizeable magnetic susceptibility and insulating character that can be easily grown by molecular beam epitaxy on various surfaces. In order to include it into the select palette of van der Waals materials used to engineer functional heterostructures, it is necessary to confirm its magnetic and electronic ground states, and understand the influence of the supporting substrate. In this work, we unambiguously demonstrate ferromagnetic ordering in a single-layer FeCl₂ on Au(111) by means of spin-polarized scanning tunnelling microscopy. The material features a relatively wide insulating gap of 3.3 eV and a strongly spin-polarized conduction band that emerges at 1.5 eV above the Fermi level. Atomic scale defects with triangular shape play a primary role in the electronic gap and spin density distribution. Specifically, in a region of 1.6 nm around each defect, the conduction band is locally suppressed and the tunnelling magnetoconductance is reduced a factor of four. By tracking the spin-dependent tunnelling conductance as a function of the applied magnetic field, we record atomically resolved hysteresis loops, revealing a soft ferromagnetic ground state with pronounced out-of-plane anisotropy and coercive fields in the range of 15–50 mT.

Keywords: Transition metal dihalide | Van der Waals materials | spin-polarized STM | 2D magnetism | structural defects

1 | Introduction

The ability of two-dimensional (2D) materials to maintain long-range magnetic ordering has significant technological implications. They can induce magnetic functionalities, such as spin filtering or spin current generation, when incorporated into van der Waals (vdW) heterostructure stacks [1]. One primary drawback is the instability of collective magnetic behaviour against thermal fluctuations in the thermodynamic limit [2]. Due to magneto-crystalline anisotropy [3–5] and finite size effects [6], a magnetic ground state can be stabilized in some compounds. Most of the few examples of magnetic two-dimensional (2D) materials are obtained by mechanical exfoliation. An exception are transition metal (TM) halides, which can be grown by molecular beam epitaxy via stoichiometric powder sublimation. This allows for a much broader range of fabrication strategies and provides greater design flexibility for atomically thin devices.

In the strict 2D limit, extended ferromagnetic order is known to occur in TM trihalides such as CrI₃ [4] and CrCl₃ [7]. TM dihalides (TMDH) would be advantageous because of their simpler stoichiometry (TMX₂). However, their magnetic properties have not yet been well established. First-principles calculations predict that the dichlorides (X=Cl) with 1T phase (TM = Fe, Co, or Ni) are ferromagnetic Mott insulators [8].

Nonetheless, the possibility of competing exchange interactions (direct exchange between the transition metal ions and superexchange through the halogen p-orbitals) and the geometrical frustration inherent to any triangular unit cell can promote other types of magnetic textures, such as antiferromagnetic arrangements or non-collinear ordering [9, 10].

Recent experimental studies have provided growing evidence of magnetic ordering in this class of materials. The most common strategy is to track the dichroic contrast stemming from the TM ions as a function of field and temperature, using either X-ray photoemission spectroscopy [11–14] or Raman scattering [15]. These are spatial-averaging techniques that cannot address individually each layer when different thicknesses coexist. In such cases, interlayer coupling effects may arise, concealing the intrinsic properties of the single-layer system. These mesoscopic techniques also fail to address the impact of point defects on the local magnetization, or the presence of non-collinear spin arrangements [11, 12] arising from geometrical frustration and competing exchange interactions. Consequently, in this potentially heterogeneous system, the observation of a hysteresis cycle or spontaneous remanence at zero field has been elusive to date. Such observation would serve as positive, unquestionable proof of long-range ferromagnetic ordering, as has been demonstrated for other two-dimensional (2D) ferromagnetic materials [3–5].

In this context, local measurement techniques become essential for understanding the magnetic character of

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TMDHs. The available magnetometry methods offering the ultimate spatial resolution are spin-polarized scanning tunnelling microscopy (SP-STM) [16] and inelastic spectroscopy of a nickelocene (NiCp₂) molecule adsorbed on the tip apex [17, 18]. The latter is a technically demanding variant of inelastic spin-flip spectroscopy, and requires computational modelling of the low energy excitations of NiCp₂ using multiple adjustable parameters [18, 19]. It has been applied to probe out-of-plane spin moments in FeCl₂ [13]. However, similar signals can be retrieved from isolated paramagnetic spins on surfaces [17, 20, 21] and the spin excitations of the NiCp₂ are insensitive to the sign of the surface magnetic moment. All this renders the NiCp₂-based inelastic spectroscopy insufficient on its own to uncover collective magnetic ordering.

Here, we resort to SP-STM to investigate the dependence of the spin density of a single FeCl₂ slab on Au(111) as a function of magnetic field. We find that the local magnetization of each FeCl₂ flake corresponds to that of a soft ferromagnet with pronounced out-of-plane magnetocrystalline anisotropy, with nearly squared magnetization loops and coercive fields of a few tens of mT. We also reveal the spectroscopic fingerprints of the tunnelling differential conductance associated to the material's intrinsic spin-polarized bands. The magnetization and density of states related to the conduction band are strongly reduced near the characteristic atomic scale defects of the FeCl₂ monolayer (ML) on Au(111). Furthermore, we investigate a set of distinct resonances appearing around the Fermi level, which could be mistaken for metallic states of FeCl₂, but are originating from the quasiparticle interference patterns of the shifted surface state of Au(111). This phenomenon is analogous to that observed in CoCl₂/Au(111) [14] and graphene on Cu(111) [22].

2 | Results and discussion

Figure 1 (a) shows an extended FeCl₂ ML grown by molecular beam epitaxy (see Supporting Fig. S1). Previous studies show that the FeCl₂ ML crystallizes in the $P\bar{3}m1$ space group, taking the so called 1T structure. This can be described as one plane of Fe atoms in a hexagonal lattice, sandwiched by two atomic planes of Cl, as shown in the top and side views of Fig. 1 (d). The Cl atoms bond to the Fe ions forming an octahedral environment. This gives rise to the splitting of the d-states into the t_{2g} and e_g sub-spaces, which determine the magnetocrystalline anisotropy depending on the electron filling of the TM ion in the high spin state (d^6 and $S = 2$ for Fe²⁺) [23]. The Cl atoms in the upper plane form a hexagonal lattice as shown in Figs. 1 (b),(c). We obtain a lattice parameter 3.59(5) Å in good agreement with existing experimental references [13, 24].

2.1 - Influence of structural defects on the electronic structure

The images in Figs. 1 (b)-(c) show some triangular defects (TD) that can only be imaged with atomic resolution at negative bias $V_b < -0.5$ V. Figure 1 (b) exemplifies the abundance of TD in the FeCl₂ ML. The length of the TD side equals to three FeCl₂ unit cells, and they display a slightly brighter contour than the surrounding atoms in topography images at $V_b = -1$ V. The yellow triangle in Fig. 1 (d) shows a sketch of the atomic sites involved in the TD. Similar defects have been reported in iso-structural vdW materials such as TiSe₂ [25] and HfTe₂ [26].

The TDs have a strong impact in the local density of states retrieved by scanning tunnelling spectroscopy. As seen in Fig. 1 (e), the differential conductance (dI/dV) spectra displays a stepped feature at 1.5 V in distant positions from TD, whereas it is absent on top them. This feature is typical of the onset of a dispersing conduction band (CB), and it has been observed in defect-less FeCl₂ grown on HOPG [27] and Bi₂Se₃ [28]. The bottom panel of Fig. 1 (e) shows dI/dV acquired in constant current mode in a large energy window, where two additional ubiquitous peaks at -1.8 V and 2.6 V are visible, in line with earlier spectroscopy studies [13]. The peak in the occupied states region marks the energy of the valence band (VB). Therefore, the energy gap of FeCl₂ ML amounts to 3.3 eV. In addition, there is a significant intensity around the Fermi level, starting at about 0.2 eV above onset of the Au(111) surface state, which will be discussed in the following section.

The coloured triangles in Figs. 1 (b),(c) indicate the positions of the TD. The simultaneous dI/dV maps taken at 1.5 V reveal the suppression of the CB contribution around the TD. We often observe a tip-induced displacement of the TD, which can be appreciated in Fig. 1 (c) in a straight comparison of the topographies before and after measuring the dI/dV map. The map becomes glitchy because of the rapid hopping of the TD during the measurement. We observed that the hopping probability of the TD increases for increasing tunnelling current or bias, with a threshold very close to precisely 1.5 V. Contrary to the behaviour of the CB, the maps of the states near Fermi level (Fig. 1 (b) middle panel) do not show any sensitivity to the defects of the FeCl₂, with an apparently arbitrary spatial distribution.

We note that TDs share some distinct characteristics of the mobile polarons found in CoCl₂ and FeCl₂ on HOPG [29, 30]: (i) hopping is triggered by injecting electrons with potential energy above the CB onset, and they are attracted by an electric field pointing from sample to tip ($V_b > 0$); (ii) the CB is definitely disrupted by them. On the other hand, they cannot be individually created or annihilated on Au(111), as occurs on HOPG [30], and there are no ring shaped features in the dI/dV maps [29]. Therefore, a possible explanation to the mobile substitutional defects would be the displacement of a virtual polaron with life time of trapped charge much shorter than the tunnelling rate of electrons, ascribed to the stronger interaction with the Au(111) metallic states, as previ-

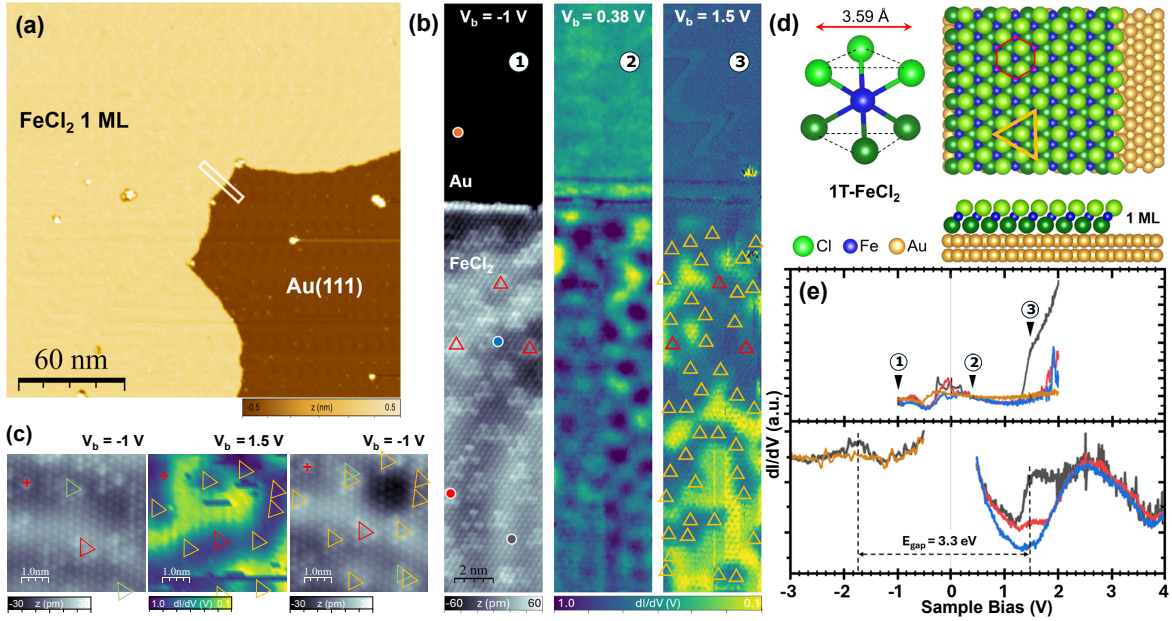


FIG. 1. **Correlation of electronic features and structural defects in FeCl₂.** (a) Survey of the FeCl₂ ML on Au(111). (b) Atomically resolved topography of the rectangular region shown in (a), together with the dI/dV maps of the same region at 0.38 V and 1.5 V. Triangles mark the position of triangular defects (TD) observed in the topography. Red triangles are a guide pointing to exactly the same position in topography and dI/dV images. (c) Atomically resolved images at $V_b = -1$ V before (left panel) and after (right panel) scanning the same area at $V_b = 1.5$ V (dI/dV map, middle panel). The red cross marks the same atomic position in all three images, and the red triangle also serves as a reference. TD are outlined by green[yellow] triangles before[after] the 1.5 V scan. The two TD singled out in the left panel shift by one lattice site in the right panel. (d) Ball model representation of a ML 1T-FeCl₂/Au(111), shown in side and top views. The red arrow indicates the unit cell. Each ML is composed of three atomic planes arranged in such a way that the Fe ion sits in an octahedral environment. (e) dI/dV spectroscopy acquired in the positions marked in (b) by dots with the same colour code. Top and bottom panel curves are taken in constant height and constant current modes, respectively. The feature associated to the conduction band (3) is only visible in sample spots without TD nearby. STM parameters: (a) $V_s/I_s = 1.3$ V/100 pA; (b) From left to right $V_s/I_s = -1$ V/100 pA, 0.38 V/70 pA, and 1.5 V/150 pA; lock-in modulation $V_{mod} = 10$ mV rms. (c) From left to right: $V_s/I_s = -1$ V/150 pA; $V_s/I_s = -1$ V/15 pA and $V_b = 1.5$ V with $V_{mod} = 10$ mV; $V_s/I_s = -1$ V/150 pA. (e) Constant height spectroscopy in top panel is initially regulated at 1 V/200 pA and $V_{mod} = 5$ mV. Regulation parameters for constant current measurements in bottom panel are 1.5 V/70 pA and -1 V/70 pA for the positive and negative bias branches respectively, and $V_{mod} = 10$ mV. Temperature is 4.3 K.

ously suggested for CoCl₂/Au(111) [29]. However, unravelling the nature of the TD requires dedicated statistical measurements and theoretical analyses that account for role of the substrate, which lies beyond the scope of this work.

2.2 - Interface state

Before addressing the magnetic response, and in order to confirm the insulating character of this material, it is crucial to clarify the origin of the states crossing the Fermi level. Their presence makes it tempting to assign metallic character to the FeCl₂ ML. Figure 2 (c) shows selected spectra taken in different FeCl₂ positions with energy resolution of ~ 1 meV. A set of pronounced peaks with strong spatial dependence emerge, reminiscent of a quantum confinement phenomenon. To investigate the origin of these states we have followed the same procedure as in the case of CoCl₂ [14]. We take constant

current dI/dV maps at varying V_b in a region enclosing FeCl₂ and some bare Au(111), as shown in Figs. 2 (a-b). Fast Fourier Transformed (FFT) images of the maps (see Fig. 2 (d)) display characteristic quasiparticle interference rings and disks, whose radius equals the maximum scattering vector, $q = 2k_{\parallel}$, with $k_{\parallel}(E)$ being the momentum of the electrons with potential energy E . Figures 2 (e) and (f) show stack plots of radial averaged profiles of the FFT images as a function of energy ($E = eV_b$), for the Au(111) and the FeCl₂ regions respectively.

The FFT analysis reveals that the FeCl₂ low energy states ($-0.1 < V_b < 1.2$ V) follow exactly the same parabolic dispersion relation as the Au(111) surface state, with identical effective mass, but a different energy onset of -0.3 eV (see Supplementary Note II). This is confirmed mesoscopically by angle-resolved photoemission (ARPES) experiments on a sub-ML coverage FeCl₂/Au(111) sample (Figs. 2 (g-f)).

The photoemission signal is dominated by a parabolic

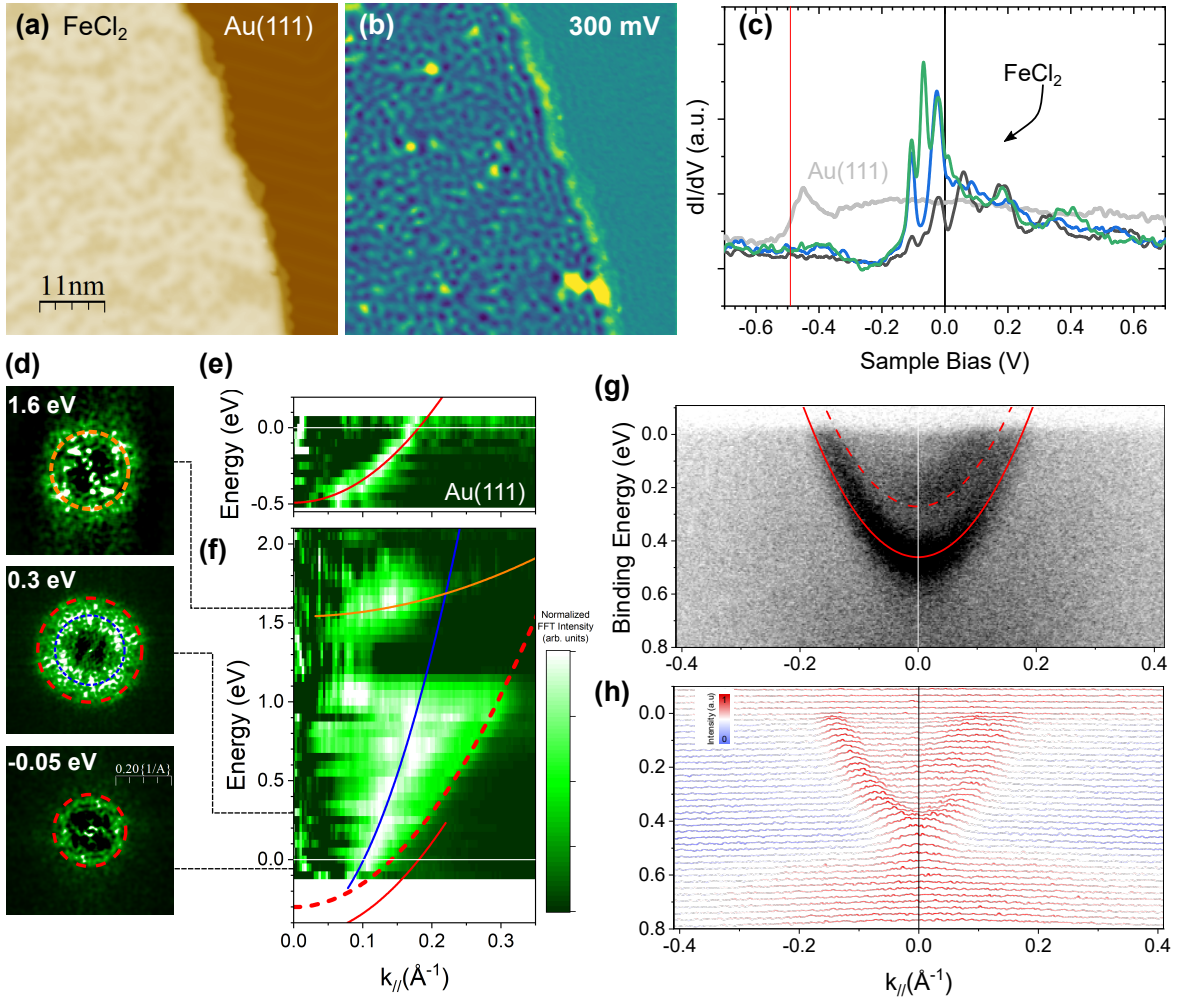


FIG. 2. **Electronic structure of FeCl₂ in reciprocal space.** (a,b) Simultaneous topography image and dI/dV map of the same region encompassing an island of FeCl₂ ML and bare Au(111). (c) High resolution dI/dV point spectroscopy of FeCl₂ and reference spectrum of Au(111) (grey). (d) Fast Fourier Transform (FFT) images [31] of the FeCl₂ region from dI/dV maps at selected energies. The mathematical processing and raw FFT images can be found in Supplementary Fig. S2. (e,f) Colour maps of the radial averaged intensities of FFT images as those in (d) as function of energy for the Au(111) and the FeCl₂ regions respectively. The red solid line is a parabolic fit to the experimental points with maximum intensity of the Au(111) radial averages, yielding an effective mass of $m^* = 0.23m_e$ [32] and energy onset of -0.49 eV (see Supplementary Note II), the latter fixed to the value marked by the red line in the Au(111) reference spectrum of (c). The dashed red line is constructed with the same m^* and a positive shift of the energy onset up to -0.30 eV. Blue and orange lines are guides to the eye with also parabolic functional form. Circles in (d) have the exact same radius as the k value of the parabolic guides with the same colour code at the indicated energies. (g) ARPES data of a sample with sub-monolayer coverage of FeCl₂ obtained at $T = 16$ K and a photon energy $h\nu = 17$ eV, overlaid with the same parabolic functions as in (e,f) shifted upwards by 30 meV. (h) Waterfall plot of the inverted ARPES intensity highlighting the two detected dispersing features. STM parameters: regulation current $I_s = 60$ pA and $V_{\text{mod}} = 30$ mV in panels (a,b) and (d,e,f); $V_s = -1$ V in topography panel (a); spectroscopy in (c) is taken with $V_s/I_s = 0.5$ V/100 pA and $V_{\text{mod}} = 2$ mV. STM temperature is 4.3 K.

band that corresponds to the surface state of the bare Au(111), with the same curvature of the dispersion relation deduced from the STM measurements (red solid line) and a 30 meV shift towards the Fermi level. This slight shift is attributed to lateral confinement effects appearing when the pristine surfaces are partially covered by thin film islands, as reported for (111) Shockley states [33]. Notably, we also find a weaker parabolic replica in ARPES maps up-shifted towards the Fermi

energy that must originate from the FeCl₂ covered regions. This feature coincides with the dispersion relation obtained in the FeCl₂ islands (red dashed line in Figs. 2 (f,g)). Therefore, we assign the highly structured dI/dV signal around the Fermi level (Fig. 2 (c)) to a modified Au(111) surface state existing at the interface between the substrate and the FeCl₂, as has been observed for graphene on Cu(111) [22]. The increase in the onset energy of the surface state is caused by the electric

field at the interface created by the electric dipoles of the FeCl_2 , which modifies the crystal electric potential at the termination of Au(111) crystal.

Additionally, a second dispersing feature marked with a blue line emerges in the energy window included in Fig. 2 (f). In the FFT images (Fig. 2 (d), middle panel) it manifests more as a ring than a disk. A free electron picture interpretation of this feature leads to a parabolic dispersion relation with much lower effective mass than the interface state. Such high-dispersive features have been reported for Au(111) and assigned to contributions of the bulk electrons [34].

Overall, this analysis establishes that the states near the Fermi level are not intrinsic to FeCl_2 and therefore neither contribute to its magnetic response nor sustain metallic behaviour.

2.3 - Spin polarization of the conduction band

Now we turn our attention to the central result of this work: the direct and quantitative measurement of the magnetic character of FeCl_2 . The high energy window of Fig. 2 (f) displays the FFT analysis of the dI/dV interference maps of the CB ($V_b > 1.5$ V). Figure 2 (d) (upper panel) shows the interference pattern derived from the dI/dV map at $V_b = 1.6$ V, exhibiting again a characteristic interference ring. This ring's radius is energy dependent, as evidenced from the additional curved feature developing at exactly 1.5 V, and highlighted by an orange line in Fig. 2 (f). This feature is compatible with a conduction band much flatter than the low energy interface state.

Spin polarized STM experiments on the FeCl_2 CB at 1.5 eV have been conducted with an out-of-plane sensitive bulk Cr-tip (see Experimental methods). The results are summarized in Fig. 3. The topography image (Fig. 3 (a)) and the dI/dV map (Fig. 3 (b)), taken under an axial field of $B = -2.7$ T, display the same fuzziness observed in Fig. 1 (c), which is a consequence of the rapid jumping of the TD when tunnelling to the CB (see Supplementary Note III). Therefore, genuine variations in the magnetic-field-dependent contrast can only be reliably probed in regions where no TD crosses the area during the acquisition of the spin-resolved images. At a large field of $B = -2.7$ T, the sample and the tip are magnetized in the same direction (see magnetic history of this experiment in Supplementary Fig. S3). Figure 3 (c) shows the dI/dV map after reversing the sample magnetization at $B = 0.09$ T, as illustrated by the sketches of tip and sample magnetic moments in (e). With the exception of the regions where mobile TDs disrupt the contrast, it is clear that the tunnelling conductance is greatly enhanced when tip and sample magnetization are antiparallel. Without loss of generality, we choose the sample's majority spin oriented towards the surface normal in this magnetic configuration.

Figures 3 (d)-(e) show selected FeCl_2 regions without significant influence of TDs displacement, together with a

schematic representation of the tip and sample magnetization in each case. The difference images (Fig. 3 (f)-(g)) between the positive and negative sample magnetization represent a map of the local spin density (see Supplementary Fig. S3 (c) for a spin density map of the entire region of interest). It is worth noticing that the dark regions at $V_b = 1.5$ V are centered at each TD (Figs. 1 (b-c)), and coincide with the regions where the spin density maps have local minima. In particular, as shown in the profiles of Fig. 3 (h)-(i), the maximum magnetic contrast in the defect free regions amounts to 40% of the average conductance, whereas at the positions of the TD the value drops significantly to $\sim 10\%$. This is still above the noise level that we retrieve from the bare Au(111) regions, which has a value of $\sim 2\%$ under our experimental conditions (see Supplementary Fig. S3 (e)). The TDs therefore suppress the spin density around them, but do not fully cancel it out.

The field dependent dI/dV conductance is plotted in Fig. 3 (j-k) for the sample spots marked in Fig. 3 (b). The spin resolved measurement captures a clear squared magnetization loop, characteristic of a soft ferromagnet with out-of-plane easy axis. The coercive field value oscillates between 15 and 50 mT. Such inhomogeneity throughout a FeCl_2 island can be attributed to domain wall pinning and inhomogeneous exchange constants in the vicinity of the TD.

Finally, Fig. 4 illustrates the energy dependence of the spin polarization in FeCl_2 . The top panel in Fig. 4 (b) shows the average dI/dV of several points far from TD marked in Fig. 4 (a) for antiparallel ($\uparrow\downarrow$) and parallel ($\downarrow\downarrow$) sample and tip magnetizations. Then the spin asymmetry is calculated as $S_a = (dI/dV_{\uparrow\downarrow} - dI/dV_{\downarrow\downarrow}) / (dI/dV_{\uparrow\downarrow} + dI/dV_{\downarrow\downarrow})$. This quantity is proportional to the sample's spin polarization. It can be seen that the conduction band ($V_b > 1.5$ V) of FeCl_2 has a strong spin polarization, whereas the in-gap region is essentially non-spin polarized. The interface states near the Fermi level show a marginal spin asymmetry of $\pm 4\%$ which is just slightly above the experimental standard deviation (blue error bars).

3 | Conclusions

By means of SP-STM we demonstrate that FeCl_2 ML on Au(111) is a soft ferromagnet with an insulating band gap of 3.3 eV and large spin polarization of the conduction band. Triangular defects centred at the position of the bottom Cl atoms interrupt locally the conduction band and induce a decrease of the spin density around them. The domain wall displacement in this material is sufficiently slow as to allow us to register different switching events in regions that are just 10 nm apart. Notably, the magnetization loops of individual FeCl_2 islands display a marked out-of-plane anisotropy character.

Our results establish the FeCl_2 monolayer as an atomically thin ferromagnetic insulator, and suggest that de-

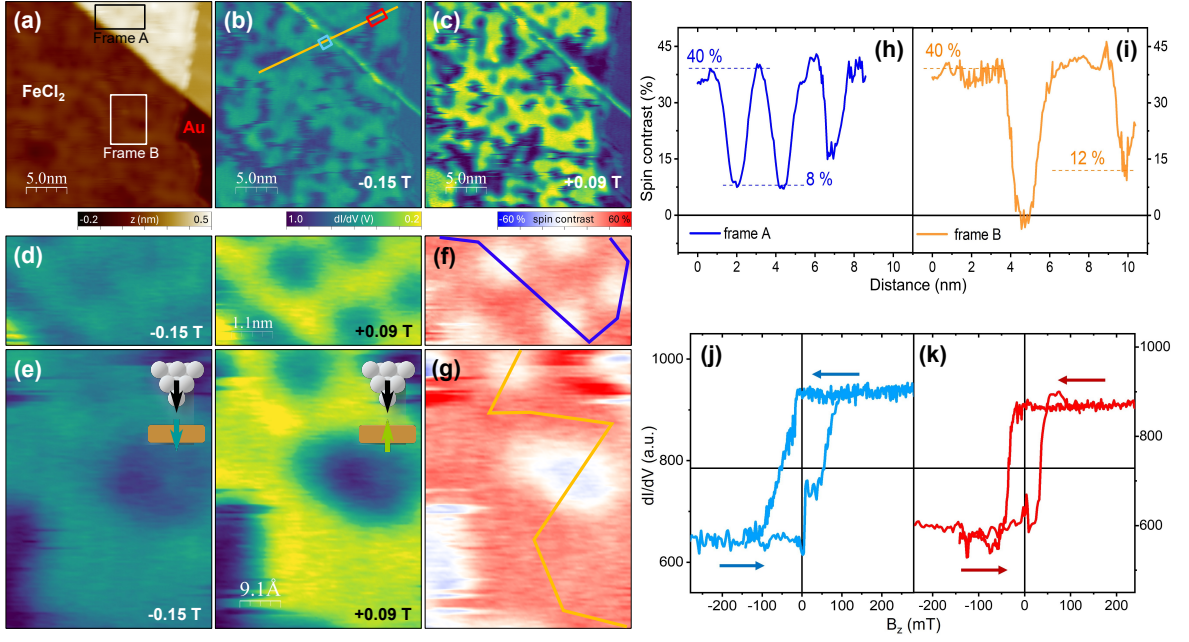


FIG. 3. Spin polarized STM of FeCl₂. (a) Topography of a ML FeCl₂ over two different Au(111) terraces. (b,c) dI/dV maps taken after ramping down the field from -2.7 T to -0.15 T ($dI/dV_{\downarrow\uparrow}$, parallel tip and sample magnetization) and further to 0.09 T ($dI/dV_{\uparrow\downarrow}$, antiparallel) after crossing zero. Dark spots occur at the position of the triangular defects (TD). See Supplemental Fig. S3 for further details on the magnetization loops. (d,e) Zooms in of frames A and B marked in (a) that enclose regions without displacement events of TD (d) during the time elapsed between both images, and with minimal displacement (e). (f,g) Spin contrast of the dI/dV (magneto-conductance) of regions A and B calculated as $(dI/dV_{\uparrow\downarrow} - dI/dV_{\downarrow\uparrow}) / ((dI/dV_{\uparrow\downarrow}) + (dI/dV_{\downarrow\uparrow}))$, where $\langle dI/dV \rangle$ is the mean differential conductance in the regions without triangular defects. (h,i) Spin contrast profiles outlined in (f,g) by the corresponding coloured lines. (j,k) Magnetic field dependence of the dI/dV contrast for a fixed tip sensitivity direction, thereby representing the local magnetization of the sample. The signal is retrieved from the blue and red areas marked by rectangles in (b), avoiding the influence of TD tip induced displacements. Note that the colour scale for each type of data has the same range throughout the panels. STM parameters: $V_s/I_s = 1.5 \text{ V}/100 \text{ pA}$, lock-in modulation 10 mV rms, $T = 1.1 \text{ K}$.

fect manipulation can be exploited in order to control spin textures and magnetic switching at the nanoscale. This work demonstrates that SP-STM is a highly suitable technique to investigate the magnetic ordering of TM halides, capable of probing the site, energy, aggregation phase and thickness specific spin density.

4 | Experimental Methods

FeCl₂ is deposited from a commercial Kentax resistive evaporator at a temperature of 350 °C to achieve a coverage of approximately 0.7 ML. The Au(111) single crystal is previously cleaned by sputtering and annealing cycles at 530 °C. To achieve the growth of the 1T crystalline phase, an optimum substrate temperature of 150 °C is required. Deviations from these parameters of dose and sample temperature result in the appearance of the well-known precursor phase of the halides [12, 35]. This phase consist in a self assembly of FeCl₂ molecules with completely different electronic and magnetic properties from the 1T monolayer slab. See Supplementary Fig. S1 for the effect of variations in substrate temperature and

coverage.

All STM measurements have been performed at the SPECS-JT-STM and the Omicron LT-qPlus of the Laboratory for Advanced Microscopy (University of Zaragoza). The whole facility operates under ultra-high-vacuum conditions ($P < 1 \times 10^{-10} \text{ mbar}$). The tip is grounded and the tunnelling bias V_b is applied to the sample. Differential tunnelling conductance dI/dV is acquired using a lock-in amplifier at a frequency of 817 or 971 Hz and r.m.s. modulation given by V_{mod} . STM images and dI/dV maps were taken either in constant height or in constant current mode, using a regulation distance determined by the set point (I_s , V_s) indicated at the corresponding caption for each data set. Image processing is performed with WSxM software package [31].

Spin averaged STM experiments are performed with electrochemically etched W tips. For spin resolved measurements, Cr-bulk tips are prepared by electrochemical etching of 1 mm wide pure Cr rods. Afterwards they are field emitted at the STM head (120 V, 1 μA , 30 min). Before proceeding to the target sample, we confirm that out-of-plane magnetic contrast is readily achieved by positive voltage pulses using the reference

signal of 2 epitaxial layers of Fe on W(110) [36]. We avoid tip-sample indentation during the whole process, and discard tips with unstable tunnelling gaps.

ARPES measurements were carried out at LOREA beamline (ALBA Synchrotron) with a base pressure of 1×10^{-10} mbar. The energy resolution (at $h\nu = 13$ eV) was ~ 15 meV and the angular resolution was $\sim 0.1^\circ$ and $\sim 0.3^\circ$ along the analyzer slit and perpendicular to it, respectively. Spectra were taken with negative circular polarisation light.

Author Contributions

A.C. and D.S. performed the spin-polarized measurements. Tunnelling spectroscopy was carried out by A.C., D.S. and E.P.S. Growth optimization was performed by E.P.S., and J.L.C.. A.J.T., S.E.H., S.K., A.S., M.C. and M.I. conducted the quasiparticle interference mapping experiments and the ARPES characterization. The project was conceived by D.S. and supervised by C.R. and D.S. The manuscript was written by D.S., A.C., J.L.C. and S.E.H. All authors contributed to the revision and discussion of the final version of the manuscript.

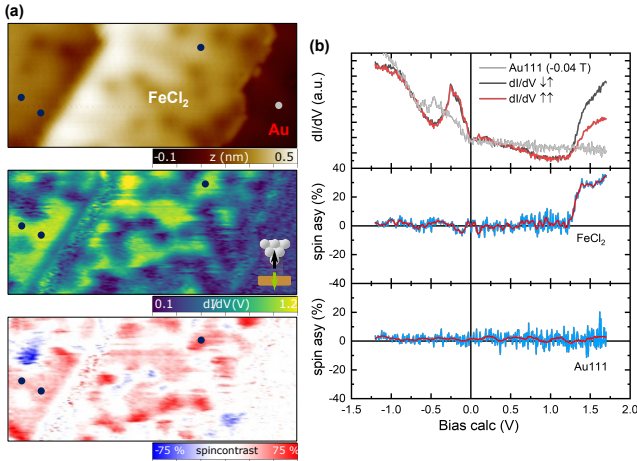


FIG. 4. Spin asymmetry as a function of energy. (a) Topography of an FeCl_2 island (top); corresponding map for antiparallel tip and sample magnetizations at -0.04 T ($dI/dV_{\downarrow\uparrow}$) (middle); and the spin contrast image (bottom) calculated by subtracting the $dI/dV_{\downarrow\uparrow}$ image above from the the $dI/dV_{\uparrow\uparrow}$ at 2.0 T. (b) Average dI/dV spectra taken at the positions marked by dark blue dots in (a), free from the influence of the TD fluctuations, for the two opposite sample magnetization directions: black spectrum corresponds to the case antiparallel to the tip ($\downarrow\uparrow$), red spectrum for the parallel case ($\uparrow\uparrow$). The middle panel shows the spin asymmetry derived from them (see main text), whereas the bottom panel shows the same quantity for the spectra taken on bare Au, which oscillates around zero reflecting the uncertainty level of this technique. Error bars are given in light blue. STM parameters: $V_s/I_s = 1.5$ V/15 pA for all images and spectroscopy curves. Lock-in modulation 8 mV rms. $T = 1.1$ K.

Acknowledgments

The authors gratefully acknowledge financial support from the Spanish MICIU/AEI/10.13039/501100011033 and by “ERDF A way of making Europe” through grants PID2022-138750NB-C21 (D.S., J.L.C.), PID2022-138750NB-C22 (M.I., C.R.) and PID2022-140845OB-C65 (M.C.) and PID2023-148225NB-C31 (C.R.), as well as from the Excellence Program Severo Ochoa CEX2023-001286-S (D.S., J.L.C.) and by the IKUR Strategy (HPC-IA and QT 2023-2025). This work has also been supported by the Aragon Government (grants E13-23R and E12-23R) and by ANPCyT-CONICET (grants PICT-INVI-00863 and PICT-2019-04545).

Conflicts of Interest

The authors declare no conflicts of interest.

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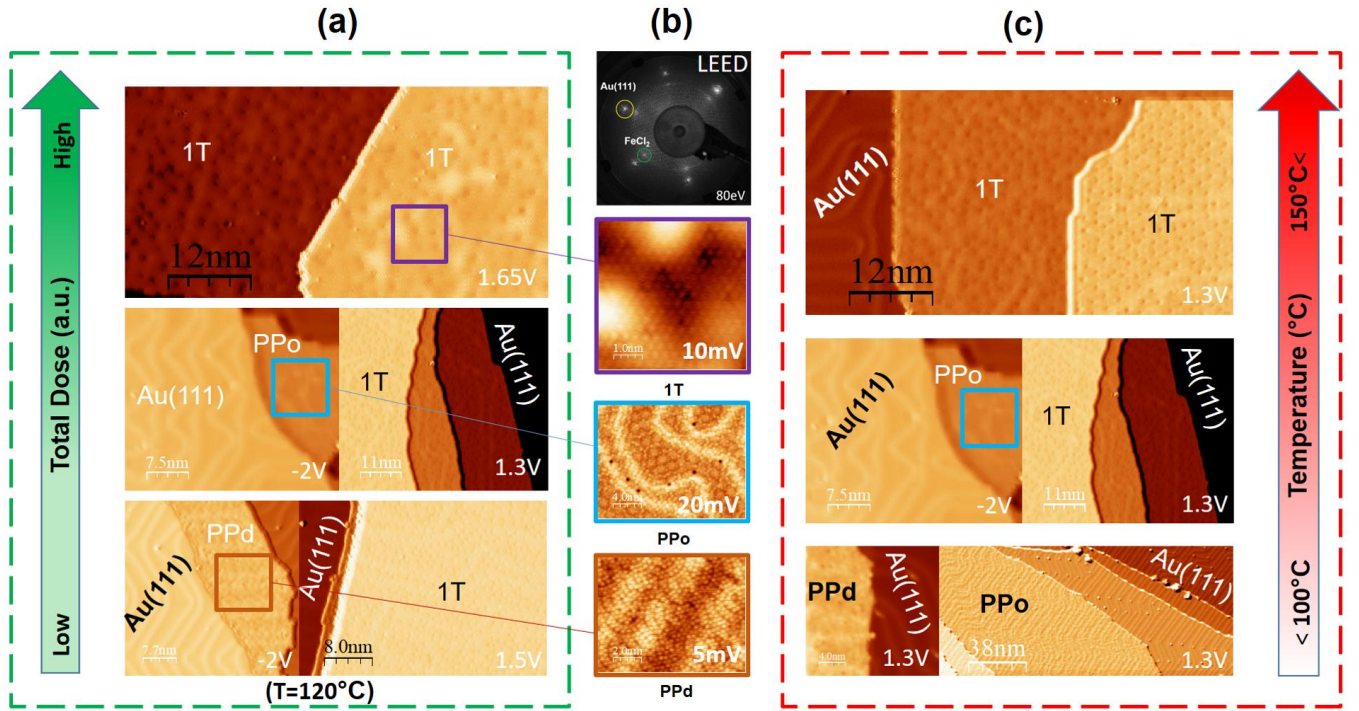
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Supporting Information

– Supplementary information for –

Atomic scale demonstration of ferromagnetism in a single layer FeCl₂ on Au(111)

1 | Extended Data on sample preparation



SUPPLEMENTARY FIGURE S1. (a) STM topography images of different samples in which the substrate is held at 120°C during the deposition of FeCl₂. The FeCl₂ dose increases in the direction indicated by the left arrow. The different phases identified in this material can be observed. (b) The top panel shows the experimental LEED pattern obtained for FeCl₂ on Au(111) ($E = 80$ eV). The coloured rectangles show atomically resolved topographies of the different phases of FeCl₂ identified in the samples shown in (a) and (c): Crystalline 1T phase (**1T**), ordered precursor phase (**PPo**) and disordered precursor phase (**PPd**). The corresponding regions in wider range images are marked in (a) and (c) with the same colour code. (c) STM topography images of different samples where the amount of deposited material is the same, whilst the substrate temperature has been increased as indicated by the red arrow on the right. The phases of FeCl₂ that appear in each case are shown. V_b for each case is given in the images.

FeCl₂ is deposited from a commercial Kentax resistive evaporator at a temperature T_{ev} previously calibrated with a quartz micro balance, at a pressure lower than 7×10^{-10} mbar. The material sublimates for $T_{ev} > 320^\circ\text{C}$ and the evaporation rate increases by one order of magnitude at $T_{ev} = 370^\circ\text{C}$. The coverage is also very sensitive to the substrate temperature during deposition, T_s . We define one ML as a complete coverage of the crystalline 1T phase shown in Fig. 1 of the main text. In the sub-ML regime, there is another competing aggregation phase: a self-assembly of FeCl₂ molecular units stabilized flat on the surface, with both Cl atoms in contact with the Au atoms, which can show from long-range ordering to just short range correlation of the intermolecular distances. This is the so-called

precursor phase (PP), previously reported for other halides like CoBr_2 [35] and FeBr_2 [12]. As shown in Fig. S1, the degree of PP ordering, and the prevalence of the PP or 1T bulk phase, is determined by the balance between total dose and T_s . In its ordered version (blue rectangle), the lattice parameter is indistinguishable by STM or LEED.

For low $T_s < 100$ °C and low coverage < 0.5 ML, the PP is preferred. If the dose is increased close to 1 ML, the material reorganizes into the 1T phase with Fe atoms in octahedral coordination with Cl ions (sketch in Fig. 1 of the main text), as shown in the Fig. S1, left panel. The right panel illustrates how the 1T phase can also be stabilized at higher temperatures. For instance, for intermediate $T_s > 120$ °C and small coverages of 0.3 to 0.7 ML, the 1T phase coexists with the ordered PP (PPo). In order to achieve an optimum 0.5 ML coverage of the 1T crystalline phase for this study, $T_s \sim 150$ °C is required. At higher temperatures $T_s = 175$ °C, only 1T phase exists for any coverage. The arch shaped LEED spots corresponding to the FeCl_2 suggest that the 1T phase has a good epitaxial registry with the Au(111) lattice directions (see Fig. S1 central panel), although there is some degree of misalignment. This is in agreement with our atomically resolved images of FeCl_2 1T slabs, where we have seldom found misalignments higher 8° , and a maximum deviation of 15° .

2 | Extended treatment of the interference patterns

Au(111)/ FeCl_2 FFT images in Fig. 2 of the article are obtained from the same dI/dV map after masking the $\text{FeCl}_2/\text{Au}(111)$ region, respectively. To better visualize the energy dependence of the FFT images utilized to compose panels (d) to (f), we have previously subtracted a radial background function $\mathcal{B}(q)$, where $q = (q_x^2 + q_y^2)^{1/2}$. $\mathcal{B}(q, E)$ is determined for each energy $E = eV_b$ as a Gaussian ($E \leq 0.95$ eV) or Lorentzian ($E \geq 1$ eV) fit to the radial average of the raw FFT image (see Fig. S2a) excluding the q region of interest, as shown by the grey shaded box in Fig. S2b. This background removes the rapidly varying noise of the images (large $q > 0.65$ nm $^{-1}$), as well as the long pitch oscillations in real space for which our 50×50 nm 2 maps do not provide enough resolution in reciprocal space (small $q < 0.15$ nm $^{-1}$). The latter is often contributed by topographic features such as the triangular defects or step edges. The final reciprocal space maps after subtraction of $\mathcal{B}(q, E)$ are shown in Fig. S2c for a few illustrative E values.

The consistency of this method can be verified by plotting the radial average of the FFT images after background removal, as done in Figs. S2d,e where we show a waterfall representation of the resulting profiles for the case of Gaussian ($E \leq 0.9$ eV) and Lorentzian ($E \geq 0.9$ eV) backgrounds, respectively. The overlap of the FFT radial profile with both methods for $E = 0.9$ eV indicates that the functional form of the background is irrelevant to the quantitative analysis in the q -range of interest. In both datasets, a clear systematic dependence of the FFT profiles on the energy can be observed. We built the color map in $E - q$ space of Fig. 2 (f) using these profiles after normalizing to their maximum value. The same procedure is applied to the Au(111) FFT patterns to compose Fig. 2 (e).

To extract the effective mass, the surface state of Au(111) and the dispersion of the interface state of FeCl_2 have been fitted with a polynomial of degree 2 over the peak position in $E - q$ space:

$$E(q) = C \cdot q^2 + B \cdot q + E_0 \quad (\text{S1})$$

with E expressed in eV and $[q] = \text{nm}^{-1}$. Then to get the effective mass, m^* , from the energy dispersion we use:

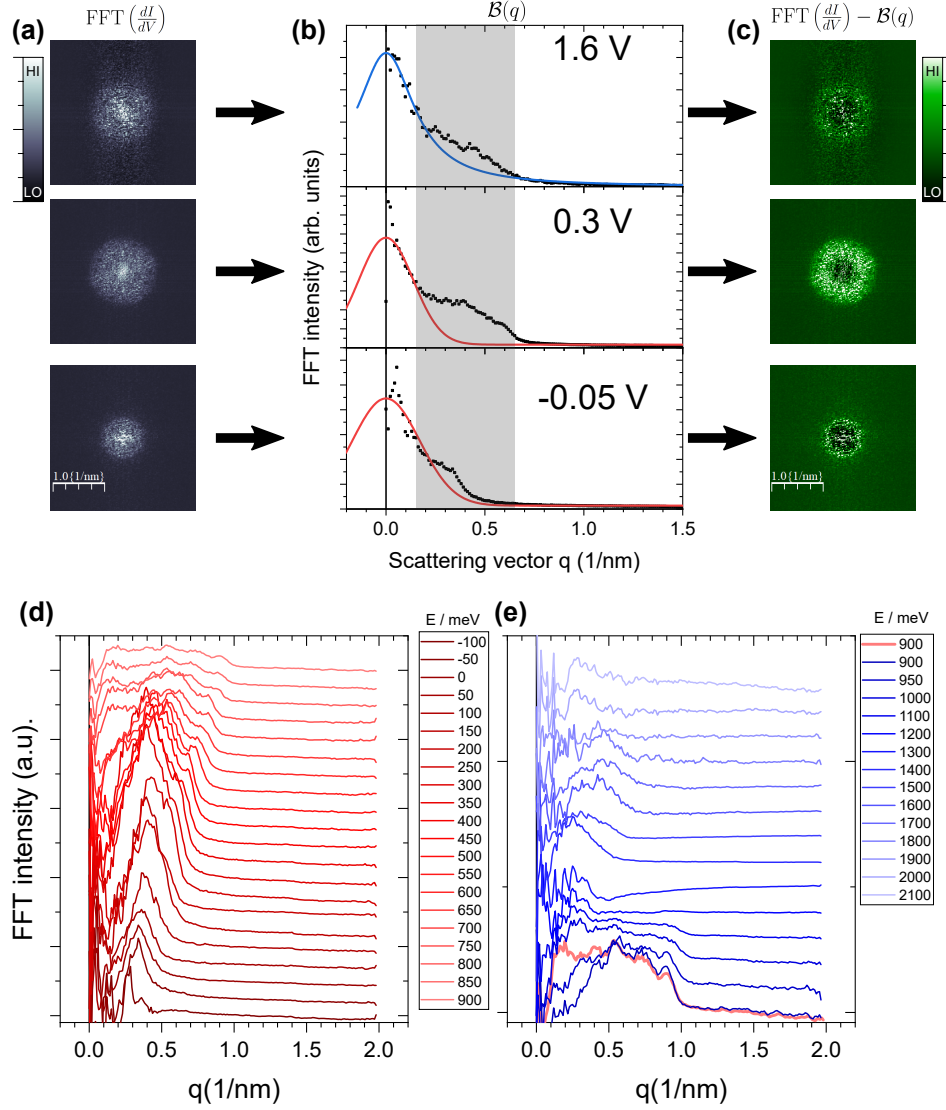
$$E(k) = \frac{\hbar^2}{2m^*} \cdot k^2 + E_0 \quad (\text{S2})$$

Since we express C in eV $\cdot \text{nm}^2$ and $2\pi q = 2k$, we have:

$$m^* = \frac{0.376}{C} m_e \quad (\text{S3})$$

where m_e is the free electron mass. The coefficients of the best minimum-squared polynomial fit to the Au(111) surface state in Fig. 2 (e) of the main article are: $C = 1.657(20)$, $B = 0$, $E_0 = -0.491$ eV. E_0 is fixed to the position of the inflection point in the dI/dV reference spectrum of the Au(111) surface state (red line in Fig. 2 (c)). The effective mass of Au(111) then yields $m^* = 0.23m_e$, in very good agreement with ref. 32. The dashed line describing the interference disk of the FeCl_2 interface state in Fig. 2 (f) is obtained by simply shifting the onset in Eq. (2) $E'_0 = E_0 + 0.19$ eV, i.e., with the same effective mass as the Au(111) surface state.

Applying the same method (Eqs. (1-3)) to the parabolic fit of the FeCl_2 conduction band (orange line in Fig. 2 (f)) we obtain $m^* = 1.26m_e$ and $E_0 = 1.54(1)$ eV.



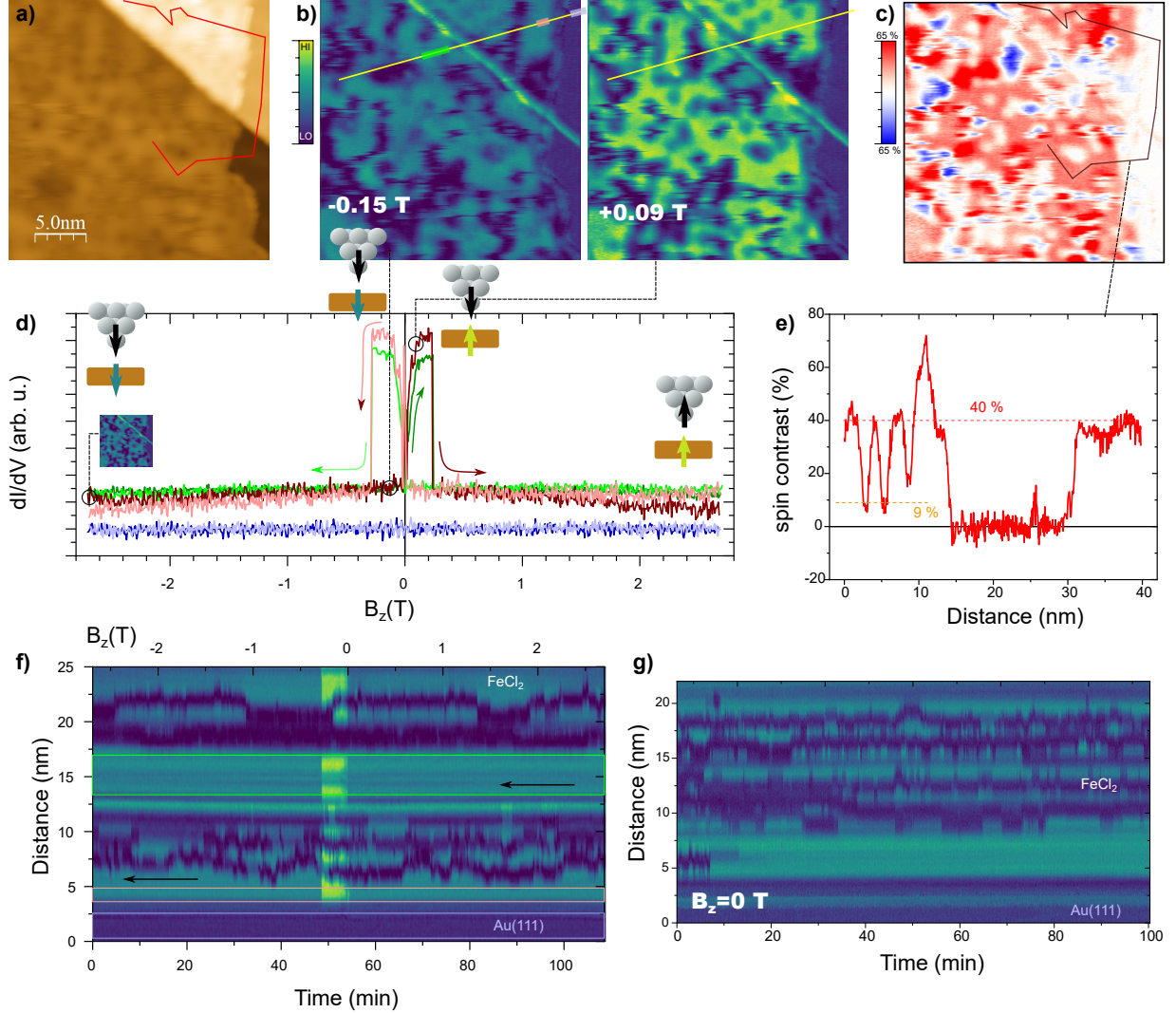
SUPPLEMENTARY FIGURE S2. **(a)** FFT images of the FeCl_2 region of dI/dV maps taken at, from top to down, $V_b = 1.6$ V, 0.3 V and -0.05 V. **(b)** The corresponding fits to Gaussian (red lines) and Lorentzian (blue lines) backgrounds of the radially averaged images in (a) excluding the shaded area from the fit. **(c)** Same as in (a) after subtracting the background functions determined as in (b). **(d,e)** Radial average of the FFT images after background removal for the energies listed in the legend. Red and blue lines represent the use of Gaussian or Lorentzian background functions, respectively.

3 | Spin-polarized measurements

We report in Figs. 3 and 4 of the article spin resolved measurements of the conduction band onset observed at $V_b = 1.5$ V. Figure S3 shows the same data set for a larger area and magnetic field range. In the spin contrast map of Fig. S3 (c,e) there are two kind of intensity levels. One is the spin density of the areas free of triangular defects (TD) with a magnetoconductance contrast of about 40%, which vanishes completely in the non-magnetic Au regions. This is the intrinsic contrast of the 1T FeCl_2 ferromagnetic phase.

There is, however, another stronger contrast which reaches maximum conductance variations of $\pm 70\%$ between dI/dV images taken at different fields. This is an artifact caused by the fluctuation of TD occurring in between the acquisition of both images. At positions where a TD appears in the last image, the disappearance of the CB onset (see main article) induces a negative (intense blue) conductance variation of around -70% . At regions where a TD drifts away to another position, the CB onset causes the opposite conductance variation of about $+70\%$ (intense red). In Figs. S3 (f), (g) we show the time dependence of such TD fluctuations. The presence of a TD manifests as a drop in the local dI/dV signal at $V_b = 1.5$ V, which jumps back and forth around the original position at slow attempt

rates between $0.1\text{--}0.001\text{ s}^{-1}$ (the largest the jump, the longer the lifetime of a TD on a particular position). The attempt rate also increases for increasing regulation I_s . The quantitative analysis of the spin contrast was performed in regions that were not visited by any TD during the measurements, as is the case of the fragments enclosed by coloured rectangles in Fig. S3 (f). It is to be noted that the TD fluctuation is not related with the magnetic field variation, as can be deduced by comparing Figs. S3 (f) and S3 (g), taken with varying and constant magnetic fields, respectively, during comparable time periods.



SUPPLEMENTARY FIGURE S3. (a) STM topography of a FeCl_2 island covering two $\text{Au}(111)$ steps. (b) dI/dV maps at $V_b = 1.5\text{ V}$ of exactly the same area for two opposite orientations of the FeCl_2 magnetization, $dI/dV_{\downarrow\downarrow}$ and $dI/dV_{\uparrow\downarrow}$ at -0.15 T and 0.09 T respectively. (c) Spin contrast map calculated as $(dI/dV_{\uparrow\downarrow} - dI/dV_{\downarrow\downarrow}) / (\langle dI/dV_{\uparrow\downarrow} \rangle + \langle dI/dV_{\downarrow\downarrow} \rangle)$, where $\langle dI/dV \rangle$ is the mean differential conductance in the regions without triangular defects. (d) dI/dV intensity at $V_b = 1.5\text{ V}$ as a function of axial magnetic field taken by continuously scanning the yellow line shown in panel (b), and extracted independently for the two different terraces of FeCl_2 (at each field value, we take the average dI/dV across the green and reddish segments on top the yellow line) and the $\text{Au}(111)$ region (blue segment). Thin coloured arrows depict the field variation direction for each one of the traces with the same colour code. Dashed lines mark the field loop status at which dI/dV maps in (b) were acquired, and the cartoons depict the corresponding tip and sample magnetic states. The small inset shows the dI/dV map at -2.7 T . (e) Spin contrast profile along the line shown in panels (a) and (c). (f) Map of stacked line profiles along the yellow line of panel (b) as a function of axial field and time. Rectangles mark the areas in which the field dependent traces of panel (d) with the same colour code are averaged. This data set corresponds to the field descent part of the loop. (g) Comparison of the same type of line scan as in panel (f) in a different island at constant zero field, showing that the jumps of the defects are not driven by the magnetic field. Panels (b), (f) and (g) share the same colour scale. STM parameters $V_s = 1.5\text{ V}$, $I_s = 100\text{ pA}$. Lock-in modulation 10 mV rms . $T = 1.1\text{ K}$.